Part 1: Investment Assets – Does the Futures Price Reflect Expected Future Spot Prices?

Consider a stock index futures contract:

$$F_0 = S_0 e^{(r-q)T}$$

where S_0 is the current spot price, (r) is the risk-free rate, and (q) is the dividend yield.

Question: Suppose investors expect the stock index to rise substantially in the future. Should the futures price rise to reflect this expectation?

Discussion: Not necessarily. The futures price is primarily determined by the **cost-of-carry relationship**, which is enforced by arbitrage.

For example, if the futures price rises substantially above its theoretical value, traders can take the arbitrage opportunity, which would push the futures price down and the spot price up until the pricing relationship is restored.

Therefore, for investment assets, **arbitrage places tight limits on how far the futures price can deviate from its theoretical cost-of-carry value**.

Market expectations can still affect futures prices indirectly, because expectations may influence today's spot price. However, expected future spot prices do not directly determine the futures price.

Transition to commodities: If arbitrage keeps investment-asset futures closely tied to their theoretical value, what happens when the underlying asset is a physical commodity that is costly or difficult to store?

Part 2: Do commodity futures prices reflect expected future spot prices?

Not necessarily. Commodity futures prices are influenced by expectations of future spot prices, but they are not simply forecasts of future spot prices.

For a storable commodity, the futures price is linked to the current spot price through the cost and benefits of carrying the commodity:

$$S_0 e^{(r+u-y)T}$$

In this equation, the expected future spot price does not appear. Therefore, it may seem that the futures price is unrelated to the expected future spot price. However, expectations about future market conditions can influence S_0, u , y. As a result, market expectations may affect the futures price indirectly, even though the expected spot price in the future does not appear explicitly in the formula.

For example, suppose the market expects **oil production to fall significantly in the future**. This expectation may cause market participants to anticipate a future shortage. As a result:

1. **The current spot price may rise** because market participants value oil more highly today.
2. **The convenience yield may rise** because having physical oil available today becomes more valuable when future supply is expected to be tight.

Therefore, expectations about future production, demand, or supply conditions can influence today's futures price (F0) indirectly through today's spot price and convenience yield, even though the futures price is not simply equal to the expected future spot price.

Part 3: Why Did Negative Oil Futures Prices During COVID-19 Not Reflect Expected Future Spot Prices?

The Event

On 20 April 2020, the May 2020 WTI crude oil futures contract settled at approximately **–$37.63 per barrel**. Did this mean the market expected oil to have a negative future value?

No.

What Happened?

COVID-19 caused a sharp collapse in oil demand as air travel, transportation, and industrial activity declined. At the same time, oil production continued, creating a severe oversupply.

Inventories accumulated rapidly, and storage capacity at Cushing, Oklahoma—the delivery location for WTI futures—was nearly exhausted.

As the May contract approached expiration, traders who did not have storage capacity or want physical delivery rushed to close their positions. Some were willing to **pay others to take the oil**, pushing the futures price into negative territory.

Why Was the Price Negative?

The negative price reflected:

- Limited storage capacity
- Imminent physical delivery
- Forced liquidation by traders
- Severe short-term market disruption

It **did not represent an expectation that oil would have a negative long-term value**. Longer-dated futures remained positive, and oil prices subsequently recovered.

The episode therefore illustrates how **short-term physical and market constraints can cause futures prices to deviate substantially from expected future spot prices**.

Question: Storage Costs Should Increase Futures Prices. Why Didn't That Happen?

According to the cost-of-carry model, higher storage costs should increase futures prices.

Key Insight

The model assumes that **storage is available at a market price**. In April 2020, this assumption broke down: storage capacity was nearly exhausted.

The problem was therefore not simply that storage became expensive—it was that **storage was becoming unavailable**.

For a trader with nowhere to store oil, taking delivery could become a liability. They might have to pay extremely high costs to find storage or transport the oil elsewhere.

Why Did the Model Break Down?

The cost-of-carry model relies on available storage and manageable physical delivery.

Key lesson: Higher storage costs increase futures prices only when storage remains available and arbitrage can function effectively.

Additional Discussion Points: Convenience Yield Became Extremely Important

Convenience yield represents the benefit of holding the physical commodity.

Normally:

- Firms holding oil inventories enjoy operational flexibility.

However, during COVID:

- Holding additional oil became a burden rather than a benefit.

One way to interpret the event is that the effective convenience yield for additional inventory became extremely negative for some market participants.